

Part A – Theory Answers

1. Mean, Median, and Mode

Mean is the average of all values, median is the middle value when data is sorted, and mode is the most frequent value. For example, if Math scores are 70, 75, 75, 80, 90 then mean = 78, median = 75, and mode = 75.

2. Variance vs Standard Deviation

Variance measures the average squared spread from the mean, while standard deviation is the square root of variance. Standard deviation is easier to interpret because it uses the original unit of the data.

3. Normal Distribution

A normal distribution is a symmetric bell-shaped distribution where most values cluster around the mean. It is commonly used to model exam scores and heights.

4. Skewness and Kurtosis

Skewness tells whether data is tilted left or right. Kurtosis measures how peaked or flat a distribution is compared to normal distribution.

5. Probability / Empirical vs Theoretical

Probability measures the chance of an event occurring. Theoretical probability is based on formulas (e.g., $1/6$ for a dice roll), while empirical probability is based on observed data or experiments.

6. Independent vs Dependent Events

Independent events do not affect each other (e.g., flipping a coin and rolling a die). Dependent events affect each other (e.g., drawing two cards without replacement).

7. Bayes Theorem Intuition

Bayes Theorem updates probability when new evidence is available. For example, if a student studies more than 5 hours, Bayes theorem helps estimate updated probability of passing.

8. Eigenvalue and Eigenvector

An eigenvector is a direction that remains unchanged after a transformation, and an eigenvalue tells how much that vector is stretched or compressed.